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Development and In Vitro Evaluation of Near-Infrared Dye-Conjugated Pullulan-Based Nanogels for M2 Macrophage-Targeted pH-Responsive Theranostic Agents

Risako Miura [1,*], Mahiro Kagami [1], Yu Kimura [1], Kazunari Akiyoshi [2] and Teruyuki Kondo [1,*]

[1] Department of Chemical Science and Engineering, Graduate School of Engineering, Kyoto University, Katsura, Nishikyo-ku, Kyoto 615-8510, Japan
[2] Institute for Integrated Cell-Material Sciences, Kyoto University, Yoshida, Sakyo-ku, Kyoto 606-8501, Japan
* Correspondence: miura.risako.8x@kyoto-u.ac.jp (R.M.); teruyuki@scl.kyoto-u.ac.jp (T.K.)

Abstract

Immunotherapy can reduce treatment-related side effects but shows limited efficacy in "cold tumors," whose immunosuppressive tumor immune microenvironment is characterized by abundant M2 macrophages and poor T cell infiltration. Because biopsy-based qualitative assessment of the tumor microenvironment is invasive and conventional imaging lacks functional information, this study aimed to develop an M2 macrophage-targeted theranostic agent enabling non-invasive photoacoustic (PA) imaging and pH-triggered cytotoxicity. A pullulan-based nanogel conjugated with mannose and near-infrared dye (IR-820) was further functionalized with the pH-responsive doxorubicin (DOX) prodrug, Aldoxorubicin, to develop Pullulan-mannose-IR820-Aldoxorubicin (PMID) nanogel. PMID was successfully synthesized, and the resulting self-assembled nanogels (<100 nm) exhibited a highly negative ζ-potential, near-infrared absorption peaks at 780 and 850 nm, and PA contrast comparable to IR-820 at 850 nm excitation. Dialysis studies demonstrated suppressed drug release at neutral pH (~20%) but accelerated release under acidic conditions, reaching ~80% within 48 h at pH 5.5, consistent with hydrazone hydrolysis and supporting tumor/lysosome-activated delivery. In RAW264.7 macrophages, PMID nanogel showed preferential uptake by M2-poralized versus M1-polarized macrophages, outperforming non-mannosylated PID nanogel and IR-820, and produced the strongest PA signal in M2 macrophage pellets. PMID nanogel also induced the highest concentration-dependent cytotoxicity in M2 macrophages, and microscopy indicated lysosomal accumulation of the nanogel with partial nuclear localization of released DOX. These findings support the use of PMID nanogel as M2 macrophage-targeted PA contrast agents and pH-responsive drug carriers with the potential to deplete immunosuppressive macrophages, modulate cold tumor microenvironments, and improve precision cancer theranostics.

Keywords: theranostics; pH-responsive release; polysaccharide; nanoparticle; photoacoustic imaging

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1. Introduction

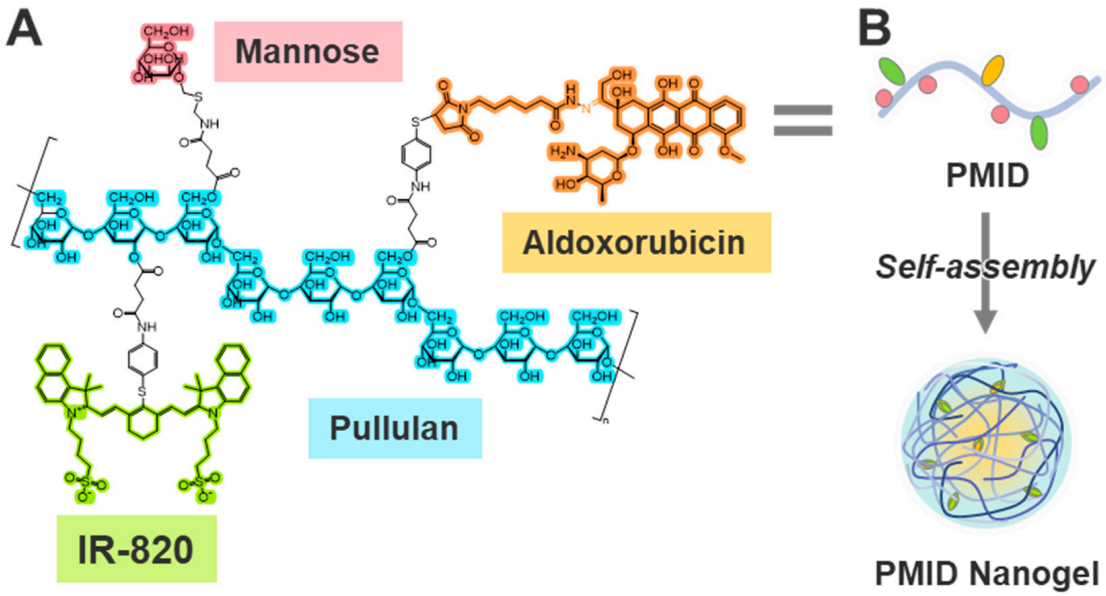
Surgery, chemotherapy, and radiation therapy are standard approaches for cancer treatment, with immunotherapy emerging as a promising alternative [1–3]. This method utilizes the patient's immune system and is anticipated to minimize the side effects. However, certain tumors, referred to as "cold tumors," exhibit a low response rate to immunotherapy [4–6]. These cold tumors are characterized by specific features in their

immune microenvironment, such as a high presence of M2 macrophages and a lack of T-cell infiltration. M2 macrophages contribute to tumor growth and metastasis by releasing Th2-related cytokines, such as IL-10 and TGF-β, which suppress cytotoxic T cell activity, and by facilitating matrix remodeling, tissue repair, and angiogenesis [7,8]. Consequently, cold tumors are prone to growth and metastasis, necessitating early diagnosis and treatment strategies.

When diagnosing the tumor microenvironment (TME), imaging of tumor morphology alone is inadequate because of insufficient qualitative data. Although biopsy is currently regarded as a reliable qualitative test for TME, it is an invasive and burdensome procedure for patients to undergo. Consequently, we have turned our attention to photoacoustic imaging (PA imaging) as a diagnostic tool, which allows for non-invasive imaging of deep body tissues [9,10]. PA imaging is a next-generation modality in which light-to-heat conversion occurs with light absorbers, such as near-infrared fluorescent dyes, under pulsed-laser irradiation.

In a previous study, we demonstrated that mannose-modified pullulan combined with a near-infrared dye (IR-820) forms a nano-sized gel (Pullulan-mannose-IR820; PMI nanogel) through the hydrophobic interaction of IR-820, serving as a PA contrast agent for colon26 tumors by specifically targeting M2 macrophages via mannose-CD206 ligand-receptor interactions [11]. Pullulan, utilized as the backbone matrix, is a highly biocompatible polysaccharide that is bioinert, non-toxic, and non-immunogenic [12,13]. Furthermore, its abundant hydroxyl groups allow for straightforward chemical modification [14]. To date, various self-assembled nanogels incorporating hydrophobic groups into pullulan have been developed, making it an exceptionally useful platform for DDS carriers [15–17]. Using pullulan as a matrix minimizes non-specific cellular interactions, thereby enabling specific cell targeting upon the modification of targeting ligands. The cell surface of M2 macrophages abundantly expresses CD206, a lectin-like domain that recognizes D-mannose and L-fucose, which has been widely utilized as a targeting site for M2 macrophages [18,19]. Therefore, the PMI nanogel was rationally designed by pendant-modifying pullulan with mannose to achieve M2 macrophage-target specificity with minimal non-specific cellular interactions and no systemic toxicity.

We hypothesized that equipping the PMI nanogel with cytotoxic agents could lead to the development of theranostic materials. In this study, we engineered a cytotoxic agent-modified PMI nanogel (Pullulan-mannose-IR820-Aldoxorubicin; PMID nanogel) to achieve M2 macrophage-targeted PA imaging and cytotoxic effects (Scheme 1). Aldoxorubicin, a prodrug of doxorubicin (DOX), is a semi-synthetic anthracycline antibiotic widely used as an anticancer drug that releases DOX through hydrolysis of the hydrazone bond in acidic environments. Although Aldoxorubicin is hydrophobic, its conjugation to the PMI nanogel enhances its hydrophilicity. Additionally, it has been reported that DOX and ICG (an analogue of IR-820) form aggregates through π-π stacking, suggesting that PMID nanogel would also form a stable structure [20]. DOX is expected to be released from the PMID nanogel upon reaching tumor sites or being internalized by cells, as both the TME and lysosomes have acidic conditions, with pH levels of 6.5 and 5.5, respectively [21–23]. This study aimed to assess the targeting specificity and cytotoxicity of the PMID nanogel towards M2 macrophages in vitro using a combination of PA imaging and controlled drug release. These results are anticipated to aid in the development of effective and precise cancer treatment strategies.



Scheme 1. Structure of PMID nanogel. (**A**) Chemical structure of Pullulan-mannose-IR820-Aldoxorubicin (PMID) and (**B**) schematic illustration of PMID nanogel. The orange bonding in Aldoxorubicin is hydrazone bonds that are hydrolyzed in an acidic environment and release DOX.

2. Materials and Methods

2.1. Materials

Allyl α-D-mannopyranoside, Irgacure2959 (4′-(2-Hydroxyethoxy)-α-hydroxy-α-methylpropiophenone), IR-820 dye (80% content), HEPES, 4-aminothiophenol, penicillin (10,000 units/mL)-streptomycin (10 mg/mL) solution, 0.25% trypsin-EDTA, and lipopolysaccharides from *Escherichia coli O55:B5* were purchased from Sigma-Aldrich (St. Louis, MO, USA). 2-aminoethanethiol, 4-dimethylaminopyridine (DMAP), and trifluoroacetic acid were purchased from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). Pullulan (M_n = 79,000) was purchased from Hayashibara Co., Ltd. (Okayama, Japan). Dulbecco's phosphate-buffered saline (PBS), succinic anhydride, triethylamine, N-hydroxysuccinimide (NHS), Triton® X-100, D(+)-glucose, sodium hydroxide, hydrochloric acid, ammonium acetate, disodium hydrogen phosphate heptahydrate, dimethyl sulfoxide, and acetonitrile were purchased from Nacalai Tesque Inc. (Kyoto, Japan). D-MEM (High Glucose) with L-Glutamine, Phenol Red and Sodium Pyruvate, Dimethyl sulfoxide (Super Dehydrated, 99.0+%), Deuterium Oxide 99.8%, 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC·HCl), tris(2-carboxyethyl)phosphine hydrochloride (TCEP·HCl), diethyl ether, D(+)-mannose, methanol, 3-methyl-1-phenyl-5-pyrazolone, chloroform and doxorubicin hydrochloride were purchased from FUJIFILM Wako Pure Chemical Corporation (Tokyo, Japan). Aldoxorubicin hydrochloride was purchased from Cayman Chemical Company (Ann Arbor, MI, USA). Sodium dihydrogen phosphate monohydrate was purchased from Merck Millipore, Inc. (Burlington, MA, USA). Fetal bovine serum was purchased from Biosera France SAS. (Boussens, France). Trypan blue stain (0.4%) was purchased from Life Technologies Corporation (Waltham, MA, USA). Cell Counting Kit-8 was purchased from DOJINDO LABORATORIES (Kumamoto, Japan). LysoTracker™ Green DND-26 was purchased from Thermo Fisher Scientific (Waltham, MA, USA). Recombinant mouse IL-4 and IFN-ɣ were purchased from BioLegend, Inc. (San Diego, CA, USA). The mouse-derived macrophage cell line, RAW264.7, was provided by the RIKEN BRC through the National Bio-Resource Project of the MEXT, Japan. The mouse fibroblast cell line, L929 (CCL-1), was purchased from the American Type Culture Collection (ATCC; Manassas, VA, USA).

2.2. Synthesis of PMID

PMID was synthesized by slightly modifying the synthesis schemes of PMI (Pullulan-Mannose-IR820), as outlined in previous studies (Schemes S1 and S2) [11]. Initially, Pullulan-COOH was synthesized by reacting the hydroxyl group of pullulan with succinic anhydride. Subsequently, Pullulan-mannose-linker (PML) was synthesized through dehydration condensation reactions involving Pullulan-COOH, NH_2-modified mannose, and 4-aminothiophenol. Finally, PMID was synthesized by mixing PML with IR-820 and Aldoxorubicin, facilitating a substitution reaction between the thiol and chloro groups and a click reaction between the thiol and maleimide groups. After each synthetic step, the polymer was precipitated and washed using diethyl ether as a poor solvent, redissolved in a minimal volume of DMSO, and subsequently dialyzed against pure water (MWCO: 12–14 kDa) to remove unreacted linkers, mannose, IR-820, and aldoxorubicin. The purified polymer was then recovered by lyophilization. PID was synthesized using the same procedure, excluding the addition of NH_2-modified mannose.

Following hydrolysis, PMP (3-Methyl-1-phenyl-5-pyrazolone) labeling was conducted, and the extent of mannose modification in PML was assessed using ultra-high-performance liquid chromatography (UHPLC; Nexera X2, SHIMADZU, Kyoto, Japan). The degree of IR-820 modification in PMID was determined by measuring the absorption in DMSO (U-3010, HITACHI, Tokyo, Japan), while the extent of Aldoxorubicin modification in PMID was assessed by measuring the fluorescence intensity in DMSO (λ_{Ex} = 485–495 nm, λ_{Em} = 595 nm; FP-6300, JASCO Corporation, Tokyo, Japan).

2.3. Preparation of PMID Nanogels

PID and PMID were dissolved in 25 mM HEPES buffer (pH 7.4) and stirred overnight. The nanogels were then sonicated using a probe-type sonicator (Q125, Qsonica, Newtown, CT, USA) at 37.5 W for 5 min, centrifuged at 6000× *g* for 20 min, and filtered through a 0.22 μm filter to remove aggregates. For the control sample, IR-820 was initially dissolved in DMSO at high concentrations (>10 mg/mL) and subsequently diluted with HEPES buffer to prevent aggregation. The hydrodynamic diameters of the Pullulan, PID nanogel and PMID nanogel were determined using dynamic light scattering (DLS; Zetasizer Nano ZS, Malvern Instruments, Worcestershire, UK). The surface electric potential of the nanogels (ζ-potential) was measured by electrophoresis light scattering (ELS; Zetasizer Nano ZS) in a disposable capillary cell.

2.4. Optical Properties

A spectrophotometer (U-3010) was used to measure the UV–Vis–NIR absorption spectra. The calibration curve was derived from the absorbance of a 1% TritonX-100 solution of IR-820 (1.25–10 μg/mL), and the IR-820 concentration in the PID nanogel and PMID nanogel were determined from its absorbance in the same solvents. A fluorescence spectrometer (FP-6300) was used to measure fluorescence spectra of IR-820, PID and PMID nanogels in HEPES Buffer or DMSO (IR-820 conc. = 10 μg/mL, λ_{Ex} = 750 nm). Photoacoustic imaging of IR-820 and PMID nanogel (IR-820 conc. = 10 μg/mL) in 1.5 mL tubes was conducted using Nexus128 (λ_{Ex} = 850 nm; Endra Life Science Inc., Ann Arbor, MI, USA). Image processing was performed on the obtained data using Osirix MD (Pixmeo SARL, Bernex, Switzerland).

In addition, the structural stability of the nanogels in HEPES buffer and DMEM containing 10% fetal bovine serum (FBS) was evaluated by monitoring their UV-Vis-NIR absorption spectra over time (1–24 h).

2.5. DOX Release

To assess the release profile of DOX, 100 μL of PMID nanogel were subjected to dialysis in either 10 mL of 0.1 M phosphate buffer (pH 5.5) or 25 mM HEPES Buffer (pH 6.5 or 7.4) at 37 °C using an Xpress Micro Dialyzer (Scienova GmbH, Jena, Germany, 3.5 kDa). The dialysate was periodically exchanged. To ensure that the DOX concentration remained above the limit of detection, the dialysate filtrates from independent trials (n = 3) were pooled and concentrated via lyophilization. Subsequently, its fluorescence intensity was measured to determine the DOX concentration (λ_{Ex} = 485–495 nm, λ_{Em} = 595 nm).

2.6. Cellular Uptake

RAW264.7 cells were cultured in high-glucose DMEM supplemented with 10% FBS and 1% penicillin-streptomycin. The cells were seeded in a 12-well plate at a density of 1.0×10^5 cells/well and incubated for 24 h. Subsequently, the medium was replaced with a culture medium containing 200 ng/mL IL-4 to induce M2 polarization. After a 24 h incubation period, IR-820, PID nanogel, and PMID nanogel were introduced (final IR-820 concentration = 4 μg/mL) and incubated for 1 h. The supernatant was then removed, and the cells were washed with DMEM and PBS before being lysed in 1% TritonX-100/PBS solution. The fluorescence intensity of the cellular lysates was measured using a spectrofluorometer (FP-6300, λ_{Ex} = 745–755 nm, λ_{Em} = 830 nm). M2 macrophage differentiation was confirmed by evaluating the upregulation of the surface marker CD206 via flow cytometry (BD LSRFortessa, BD Biosciences, Franklin Lakes, NJ, USA). Additionally, cellular uptake by M1 macrophages, polarized by the addition of 20 ng/mL IFN-γ and 100 ng/mL LPS instead of IL-4, was evaluated using the same procedure. The fluorescence intensity was normalized to that of the IR-820.

2.7. PA Imaging of Cell Pellets

M2-polarized RAW264.7 cells were treated with IR-820, PID nanogel, and PMID nanogel for 1 h in a 12-well plate, followed by washing with PBS and collection using a cell scraper. The cell suspension was transferred to 1.5 mL tubes, and cell pellets were obtained by centrifugation at 1000× g for 2 min. PA imaging was conducted on the tubes containing the cell pellets (λ_{Ex} = 850 nm).

2.8. Cytotoxicity Assay

RAW264.7 cells were initially seeded in a T25 flask with medium at a density of 1.0×10^6 cells and incubated for 24 h. Subsequently, the medium was replaced with culture medium containing 200 ng/mL IL-4 to induce M2 polarization, followed by an additional 24 h incubation. The cells were then harvested and seeded into a 96-well plate at a density of 3.0×10^3 cells per well. After a 24 h pre-culture period, the culture medium was substituted with fresh medium containing varying concentrations of DOX, PID, and PMID nanogels. Cell viability was assessed after a 48 h incubation using Cell Count Reagent SF. As control, the cytotoxicity was evaluated against M1 polarized RAW264.7 cells and L929 cells (mouse fibroblast-like cell line) were evaluated in the same procedure.

2.9. Hemolysis Assay

CDF1 mice were purchased from Shimizu Laboratory Supplies Co., Ltd. (Kyoto, Japan). All animal experiments were approved by the Departmental Committee of Animal Experimentation at Kyoto University (Approval No. 2026-14, 30 March 2026) and conducted in accordance with the Regulations on Animal Experimentation at Kyoto University.

Whole blood was collected from CDF1 mice into EDTA-treated tubes, and red blood cells (RBCs) were isolated and washed with PBS by centrifugation (800× g, 5 min). RBC

suspensions were incubated with IR-820, PID nanogel, and PMID nanogel at IR-820 conc. of 10 μg/mL at 37 °C for 1 h. After centrifugation (1000× *g*, 5 min), the absorbance of the supernatant was measured at 540 nm. PBS and 0.1% TritonX-100 solutions were used as negative (0% hemolysis) and positive (100% hemolysis) controls, respectively.

2.10. Fluorescence Microscopy

RAW264.7 cells were seeded into glass-bottomed chambers at a density of 1.5×10^4 cells per well. Following a 24 h pre-culture, the medium was replaced with culture medium containing 200 ng/mL IL-4 to induce M2 polarization, and the cells were incubated for an additional 24 h. PMID nanogel were then added to achieve a final IR-820 concentration of 20 μg/mL (DOX concentration of 0.57 μg/mL). After 4 h, lysosomal staining was performed using Lysotracker. The fluorescence of Lysotracker, IR-820, and DOX in M2-polarized RAW264.7 cells was observed using a fluorescence microscope (BZ-810; KEYENCE, Osaka, Japan). The colocalization efficiencies among lysosomes, IR-820, and DOX were evaluated by calculating Pearson's *R* values using Fiji (ImageJ 1.54p; National Institutes of Health, Bethesda, MD, USA).

2.11. Statistical Analysis

Statistical analyses were performed using OriginPro 2025 (OriginLab Corporation, Northampton, MA, USA) and GraphPad Prism 9 (GraphPad Software, La Jolla, CA, USA). Data are presented as mean ± S.D. from at least three independent experiments. Statistical significance was determined using analysis of variance (ANOVA) followed by Tukey's post hoc test. All *p*-values were two-sided, with *p*-values less than 0.05 considered significant.

3. Results

3.1. Synthesis of PMID

The ^{1}H NMR spectrum of Pullulan-COOH in D_2O exhibited several distinctive peaks, notably the anomeric protons of pullulan at 5.78 and 5.37 ppm, along with the protons attributed to succinic acid at 2.92 ppm (Figure S1). The modification ratio of the carboxyl group was determined by integrating the areas of the ^{1}H NMR peaks, resulting in 30 carboxyl groups per 100 glucose units. The degree of mannose modification was assessed by analyzing the monosaccharide composition of PML using UHPLC, which yielded 22 mannose units per 100 glucose units (Figure S2). Additionally, the absorbance of IR-820 and the fluorescence intensity of Aldoxorubicin in PMID dissolved in DMSO were measured, determining the degrees of IR-820 and Aldoxorubicin modifications as 2.9 and 0.12, respectively, per 100 glucose units.

3.2. Characterization of PMID Nanogel

The diameters and ζ-potentials of the pullulan, PID nanogel and PMID nanogel were measured using dynamic light scattering (DLS) and electrophoretic light scattering (ELS), and the results are presented in Table 1. The diameter of the PID nanogel and PMID nanogel were larger than that of pullulan, indicating the formation of self-assembled nanogels through hydrophobic interactions between IR-820 and Aldoxorubicin. The polydispersity index (PDI) values were determined to be 0.18 for native pullulan, 0.54 for PID nanogels, and 0.41 for PMID nanogels, indicating that the prepared nanogels possessed moderate monodispersity. Regarding the ζ-potentials, the PID nanogel and PMID nanogel exhibited a highly negative charge, primarily due to the presence of sulfonic acid groups in IR-820.

Table 1. Diameter and ζ-potential of pullulan and PMID nanogel.

	Diameter (nm)	ζ-Potential (mV)
Pullulan	20.0 ± 0.5	−18.8 ± 0.4
PID nanogel	36.1 ± 0.6	−31.6 ± 1.4
PMID nanogel	79.2 ± 0.6	−46.5 ± 1.1

The UV–Vis–NIR absorption spectra of IR-820, PID nanogel and PMID nanogel were measured. Both PMID nanogel and IR-820 exhibited similar absorption characteristics, with peaks at 780 nm and 850 nm in HEPES buffer, as illustrated in Figure 1A. PID nanogel also exhibited a broad peak around 730 nm. In the presence of 1% TritonX-100, the absorption peak of PID nanogel and PMID nanogel became more defined and was observed at 850 nm, closely resembling the absorption profile of IR-820 in DMSO and 1% TritonX-100 (Figure 1B). Fluorescence spectral measurements revealed that IR-820, PID nanogel, and PMID nanogel exhibited almost no fluorescence in HEPES buffer, indicating the occurrence of aggregation-caused quenching (ACQ) induced by the hydrophobic association of IR-820 molecules (Figure S3A). Conversely, under conditions where the assembly of IR-820 was completely disrupted (in DMSO), a distinct recovery of fluorescence was observed (Figure S3B).

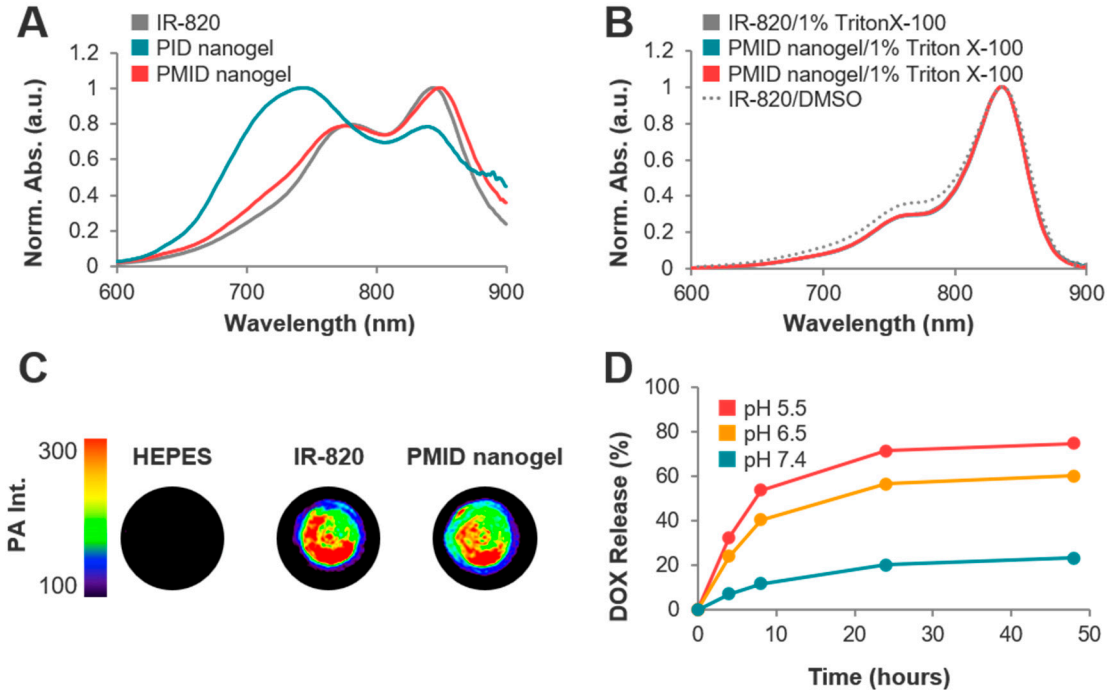


Figure 1. Characterization of PMID nanogel. Absorption spectra of IR-820, PID nanogel and PMID nanogel in (**A**) HEPES buffer, (**B**) 1% TritonX-100 or DMSO. (**C**) PA imaging of HEPES buffer, IR-820 and PMID nanogel (IR-820 conc. = 10 μg/mL, λ_{Ex} = 850 nm). (**D**) Time-dependent DOX release from PMID nanogel in different pH buffer conditions. Data are presented as mean values from pooled and concentrated dialysates (n = 3 independent trials per time point) to ensure analytical sensitivity.

Photoacoustic (PA) imaging of the PMID nanogel was conducted, as depicted in Figure 1C. The excitation wavelength was 850 nm, which corresponds to the peak of the absorption spectrum. IR-820 has been extensively researched as a PA contrast agent, similar to ICG [24,25], and the PMID nanogel demonstrated PA contrast comparable to that of IR-820.

Tracking the temporal changes in the UV-Vis-NIR absorption spectra in HEPES buffer and DMEM revealed no detectable spectral changes in either medium at 4 °C. In contrast,

upon incubation in DMEM at 37 °C, a distinct spectral shift was observed after 24 h, converging into a profile similar to that obtained with 1% Triton X-100 treatment (Figure S4).

The PMID nanogel underwent dialysis against three distinct pH buffers (pH 5.5, 6.5, and 7.4), and the fluorescence intensity of the dialysates was measured to assess the DOX release profile of the PMID nanogel (Figure 1D). The pH of 5.5 corresponds to the lysosomal environment, while the pH of 6.5 is representative of the tumor microenvironment. DOX exhibited a more rapid release under acidic conditions than under neutral conditions, with the final release amounting to nearly 80% within 48 h at pH 5.5.

3.3. Interaction with Macrophages

The fluorescence intensities of IR-820, PID nanogel, and PMID nanogel-treated macrophage lysates were measured to assess cellular uptake. The fluorescence intensity was normalized to that of the IR-820-treated cell lysates. In M1 macrophages, IR-820 alone exhibited the highest fluorescence intensity, indicating the greatest uptake efficiency (Figure 2A). The fluorescence intensities of PID nanogel and PMID nanogel were nearly identical. Conversely, when M2 macrophages were treated, the trend differed, with PMID nanogel demonstrating the highest fluorescence intensity (Figure 2B). When the incubation time for M2 macrophages was extended to 4 h, PMID nanogel consistently maintained the highest cellular uptake; however, the difference in fluorescence intensity between PID and PMID nanogels became less pronounced (Figure S5).

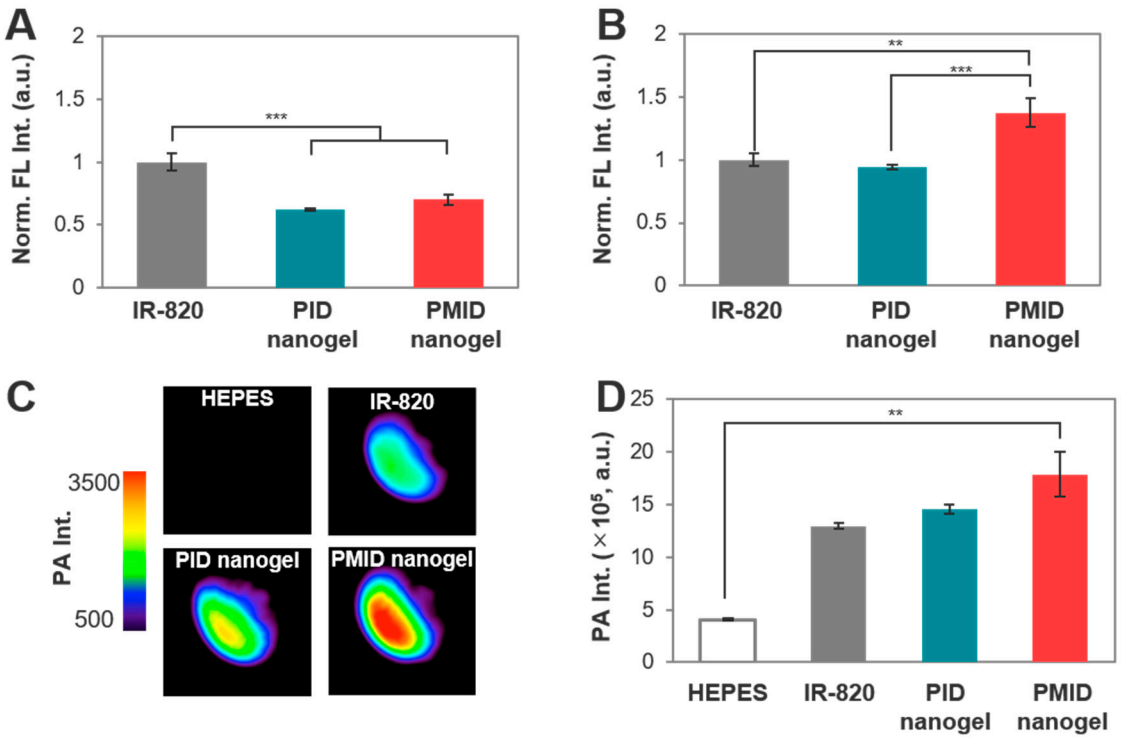


Figure 2. Interaction with macrophages. The uptake of IR-820, PID nanogel and PMID nanogel by (**A**) M1 macrophages and (**B**) M2 macrophages determined by the fluorescence intensity of IR-820 and normalized with the intensity. (**C**) PA imaging of sample treated M2 macrophage pellets (λ_{Ex} = 850 nm) and (**D**) the PA intensity. (** $p < 0.01$, *** $p < 0.001$).

Following the interaction of IR-820, PID nanogel, and PMID nanogel with M2 macrophages, photoacoustic (PA) imaging was performed on the cell pellets (Figure 2C). PA imaging and subsequent intensity analysis revealed that the PMID nanogel provided the highest contrast for visualizing M2 macrophages (Figure 2D).

The cytotoxic effects of DOX, PID nanogel, and PMID nanogel on M2 macrophages were assessed (Figure 3). Each sample exhibited cytotoxicity in a concentration-dependent manner. Notably, the PMID nanogel exhibited the highest efficiency at each concentration level, consistent with the cellular uptake results. On the other hand, against M1 macrophages and normal L929 fibroblasts, high cell viability exceeding 90% was maintained at concentrations of 0.08 μg/mL and below. Although statistically significant differences were observed at certain concentration points, no consistent toxicity trend was detected among the different samples (Figure S6). Furthermore, a hemolysis assay revealed that IR-820, PID nanogel, and PMID nanogel all exhibited a very low hemolytic ratio of approximately 2%, demonstrating their high hemocompatibility (Figure S7).

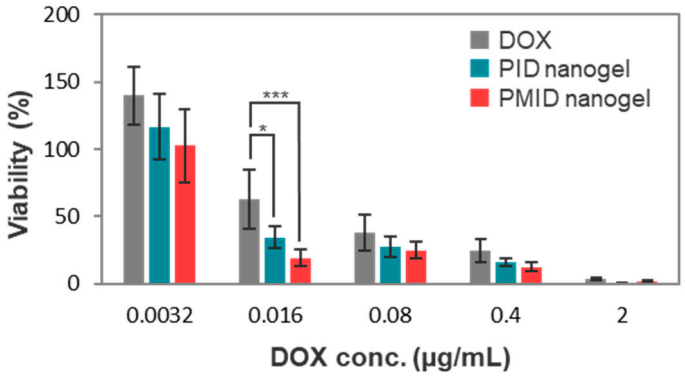


Figure 3. Cytotoxicity assay. M2 macrophages were treated with DOX, PID nanogel and PMID nanogel, followed by cellular viability test. (* $p < 0.05$, *** $p < 0.001$).

Fluorescence microscopy was used to assess the intracellular localization of PMID nanogel within M2 macrophages (Figure 4). Based on the line profile analysis of fluorescence intensity across the cells and the comparison of Pearson's *R* values, the fluorescence signals of IR-820 and Lysotracker exhibited co-localization (Figure 4B,C). In contrast, the fluorescence of DOX (Aldoxorubicin) demonstrated partial co-localization with Lysotracker and was also distributed in regions where neither lysosomes nor IR-820 were present, presumably corresponding to the cell nuclei.

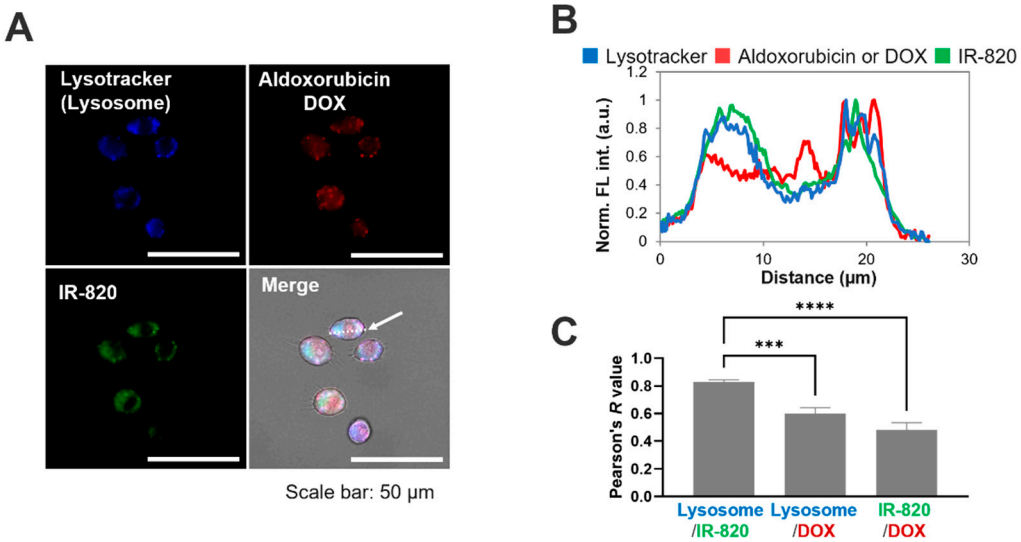


Figure 4. Intracellular distribution observation of PMID nanogel and DOX. (**A**) Fluorescence microscopy imaging of PMID nanogel-treated M2 macrophages and (**B**) the normalized fluorescence intensity on the dashed line with the white arrow. (**C**) Evaluation of colocalization between individual components (Pearson's *R* value, *** $p < 0.001$, **** $p < 0.0001$).

4. Discussion

The successful synthesis of PMID was confirmed by evaluating each modification component. Regarding the mannose modification of Pullulan-COOH, because 30 COOH groups were introduced per 100 monosaccharide units, the molecular weight of Pullulan-COOH was calculated to be approximately 120,000 g/mol. Using the molecular weights, it was calculated that IR-820 and Aldoxorubicin were conjugated to 14 and 0.58 molecules per PMID molecule, respectively. DLS analysis indicated that PMID formed self-assembled nanogels in water via hydrophobic interactions between IR-820 and Aldoxorubicin. Although each single polymer chain carries an average of 0.58 Aldoxorubicin molecules, these intermolecular hydrophobic interactions drive the aggregation of multiple polymer chains into a single nanogel particle, ensuring that each nanogel particle contains multiple Aldoxorubicin molecules. It has been reported that ICG, a structural analog of IR-820, and DOX form aggregates [13], suggesting that both molecules are associated with the physical cross-linking points of PMID nanogels. The conjugation ratio of Aldoxorubicin (0.12 per 100 glucose units) was selected to optimize the hydrophilic/hydrophobic balance for colloidal stability, as higher hydrophobic modifications were found to cause gradual self-aggregation over time in our preliminary studies. Furthermore, because PMID nanogels combine chemotherapy with IR-820-mediated photothermal therapy (PTT), the synergistic therapeutic effect compensates for the lower drug payload. For future applications requiring a higher DOX payload, the conjugation ratio could be improved by increasing hydrophilicity—such as by introducing short PEG chains or using alternative polysaccharide backbones—to prevent hydrophobic aggregation while maintaining dispersion stability.

The physical properties of PMID nanogel are conducive to bioimaging applications. Compared to pullulan, the diameter of PMID nanogel increased, indicating the formation of nanogels. This is thought to be induced by hydrophobic interactions between IR-820 molecules. The particle size of the PID nanogel was similarly larger than that of pullulan, but much smaller than that of the PMID nanogel. This is thought to be because, in the PID nanogel, mannose is not modified, resulting in a balance of hydrophilicity and hydrophobicity that is shifted toward hydrophobicity compared to the PMID nanogel, thereby enhancing hydrophobic interactions and leading to more compact packing. The diameter of PID nanogel and PMID nanogel were less than 100 nm, and they possessed negatively charged ζ-potentials. These characteristics suggest that PID nanogel and PMID nanogel can evade non-specific interactions with proteins and cells, thereby facilitating tumor accumulation through the enhanced permeability and retention (EPR) effect without obstructing blood vessels under biological conditions [26–28].

In the UV–Vis–NIR spectral measurements, the peak observed at 850 nm in HEPES buffer, 1% TritonX-100, and DMSO was considered to originate from the absorption of IR-820 monomers. In contrast, the pronounced peak at 780 nm observed in the HEPES buffer is thought to be attributable to IR-820 aggregates. The term "biological window" denotes the wavelength range of 650–900 nm, which is characterized by high light transmittance and minimal absorption by hemoglobin and water [29–31]. PMID nanogels demonstrate light absorption within the NIR region, which falls within this biological window, and provide photoacoustic (PA) contrast when exposed to an NIR pulsed laser. These properties render them suitable as contrast agents for in vivo imaging applications. PID nanogel exhibits a unique broad peak around 730 nm, which is considered to be an absorption derived from the *H*-aggregate dimer of IR-820 [32].

Evaluation of nanogel structural stability via temporal UV-Vis-NIR absorption measurements demonstrated no significant spectral changes in either HEPES buffer or DMEM at 4 °C, indicating that the nanogel structure remained fully intact and capable of long-term

storage under refrigerated conditions. Conversely, only when incubated in DMEM at 37 °C did the absorption peak gradually shift, eventually mirroring the profile observed in the presence of 1% Triton X-100. This structural transition suggests that interaction with serum proteins in DMEM disrupts the assembly of IR-820, thereby triggering the disassembly of the nanogel architecture.

Under neutral pH conditions, the release of DOX from the PMID nanogel was significantly suppressed (approximately 20%). Conversely, under acidic conditions, the release of DOX was markedly accelerated, reaching approximately 80%, with a more rapid release observed at lower pH levels. These findings indicate that hydrolysis of the hydrazone group of Aldoxorubicin occurred within the PMID nanogel. The pH-responsive release of DOX from this PMID nanogel is anticipated to facilitate DOX release following uptake into the tumor microenvironment or M2 macrophages, potentially mitigating the systemic side effects associated with the administration of DOX alone.

The uptake of nanogels by macrophages was assessed by measuring the fluorescence intensity of IR-820. Since the cells were lysed using a high concentration of surfactant, the internalized IR-820 and nanogels were fully disassembled, thereby eliminating ACQ of IR-820. Consequently, the resulting fluorescence intensity directly corresponded to the quantitative uptake of IR-820 and nanogels. M1 macrophages exhibited higher uptake of IR-820 than PID nanogel and PMID nanogel. This is likely due to the aggregation of IR-820 in the culture medium, as determined by DLS (Figure S8), facilitating the phagocytosis of larger particles by macrophages [33,34]. In contrast, the uptake of PMID nanogel by M2 macrophages was greater than that of IR-820 and PID nanogels, indicating an active interaction between the mannose on the PMID nanogel and the CD206 receptor expressed on M2 macrophages. Indeed, flow cytometric analysis confirmed that IL-4 treatment induced macrophage polarization toward the M2 phenotype, accompanied by the upregulation of CD206 expression (Figure S9). In Addition, CD206 recognizes not only mannose but also the equatorial 3- and 4-OH groups of glucose in pullulan [35,36], resulting in increased uptake of PID nanogel by M2 macrophages compared with M1 macrophages. When the incubation time was extended to 4 h, PMID nanogel consistently exhibited the highest uptake; however, the differences relative to the other samples narrowed. This suggests that receptor–ligand mediated uptake occurs rapidly, whereas subsequent non-specific uptake gradually proceeds over time, thereby attenuating the overall uptake specificity. Moreover, photoacoustic (PA) images and the intensities of the cell pellets were consistent with the uptake levels of IR-820 or nanogels. These findings demonstrate that the PMID nanogel actively interacts with M2 macrophages and functions efficiently as a PA contrast agent.

The uptake efficiency also influences the cytotoxicity of M2 macrophages. A cytotoxicity assay revealed that the PMID nanogel exhibited greater cytotoxicity than DOX alone and the PID nanogel, suggesting that the PMID nanogel delivered a higher concentration of DOX to M2 macrophages. Notably, the cell viability exceeding 100% observed at low DOX concentrations can be attributed to hormesis, an adaptive stress-response phenomenon, and enhanced mitochondrial dehydrogenase activity induced by mild cytotoxic stress [37,38]. Although no significant difference in cytotoxicity toward M2 macrophages was observed between PID and PMID nanogels in this in vitro assay, this is likely because the prolonged 48 h incubation time led to non-specific cellular uptake, thereby diminishing the distinction between the two groups. Nevertheless, our previous study demonstrated that mannose-modified PMI nanogel markedly enhanced tumor accumulation in vivo. Therefore, PMID nanogel are expected to synergistically deplete M2 macrophages residing in the tumor microenvironment. On the other hand, low cytotoxicity was observed against M1 macrophages and L929 fibroblasts, suggesting the feasibility of selective depletion of

M2 macrophages. The relatively high cell viability maintained even with free DOX in these cell types can be attributed to differences in DOX sensitivity among different cell lines. Furthermore, the hemolysis assay demonstrated high hemocompatibility across all samples, confirming extremely low toxicity toward normal cells and blood components.

Following delivery to the cells, DOX was released from the PMID nanogel and migrated to the nucleus, where it exerted its function, as demonstrated by fluorescence microscopy. The co-localization of IR-820 with lysosomes indicated that the PMID nanogel was internalized via the endocytic pathway. Consequently, the pH-responsive linker of Aldoxorubicin in the PMID nanogel was cleaved owing to the acidic conditions within the lysosomes. M2 macrophages contribute to the formation of "cold tumors" by producing immune-suppressive cytokines; thus, their depletion can activate the immune system by mitigating immune-suppressive activities [39,40]. Recent studies have reported that reducing the number of M2 macrophages and inhibiting their function can effectively suppress tumor growth by decreasing tumor angiogenesis and enhancing immune surveillance [41,42]. Therefore, the PMID nanogel is anticipated to not only directly damage cancer cells through DOX release within the tumor microenvironment but also modulate the tumor immune microenvironment by targeting and depleting M2 macrophages, thereby promoting a state more favorable to therapeutic efficacy.

The study illustrates the targeting capability of PMID nanogel towards M2 macrophages through mannose modification, as well as its photoacoustic imaging properties and cytotoxic effects in vitro. The PMID nanogel exhibits significant theranostic potential by integrating tumor and M2 macrophage-targeted doxorubicin (DOX) release, modulating the tumor immune microenvironment, and its application as a photoacoustic contrast agent.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/jnt7030020/s1, Scheme S1: Synthesis of NH_2-modified mannose; Scheme S2: Synthesis of PMID; Figure S1: ^{1}H NMR spectrum of Pullulan-COOH. The degree of COOH modification was determined to be 30 per 100 glucose units from the integrated areas of the peaks (a and b; anomeric protons of pullulan, c; succinic acid); Figure S2: UHPLC analysis of hydrolyzed PML. From the areas of the peaks, the monosaccharide composition of PML was determined to be 22 mannose units per 100 glucose units; Figure S3: Fluorescence spectra of IR-820 formulations. Fluorescence spectra of IR-820, PID nanogel, and PMID nanogel measured in (A) HEPES buffer and (B) DMSO ([IR-820] = 10 μg/mL, λ_{Ex} = 750 nm). The results demonstrate that fluorescence was quenched in HEPES buffer due to ACQ effect, whereas fluorescence was fully recovered in DMSO upon disassembly of the nanogel structure. Figure S4: Stability evaluation of nanogels. Temporal UV-Vis-NIR absorption spectra of free IR-820, PID nanogels, and PMID nanogels incubated in HEPES buffer or DMEM containing 10% FBS at 4 °C or 37 °C. Under storage at 4 °C, no distinct spectral changes were observed, confirming high structural stability. In contrast, incubation in DMEM at 37 °C resulted in spectral shifts, suggesting the disassembly of the nanogel structure triggered by interactions with serum proteins; Figure S5: Evaluation of sample uptake by M2 macrophages after 4 h of interaction. As observed at 1 h, the PMID nanogel displayed the highest uptake; however, there was no significant difference among the samples, suggesting that non-specific interactions gradually occurred over time; Figure S6: Cytotoxicity evaluation against M1 macrophages and L929 fibroblasts. Cell viability of (A) M1 macrophages and (B) L929 fibroblasts treated with free DOX, PID nanogels, or PMID nanogels. Although statistically significant differences were observed at certain concentrations, no consistent toxicity trend was identified among the samples, and high cell viability exceeding 90% was maintained at DOX-equivalent concentrations of 0.08 μg/mL and below; Figure S7: Hemolysis assay. RBCs were incubated with free IR-820, PID nanogel, or PMID nanogel at 37 °C for 1 h. All samples exhibited a low hemolytic ratio of approximately 2%, demonstrating high hemocompatibility; Figure S8: Evaluation of aggregation in DMEM. Size histograms of culture medium (DMEM+10%FBS), IR-820, and PMID nanogels in DMEM, determined using DLS. The peaks ranging from a few nanometers to 100 nm originate from DMEM and FBS, which overlap with and

obscure the signals corresponding to monomeric IR-820 and PMID nanogels. Only IR-820 in DMEM showed large aggregates (red square), while the PMID nanogel did not, suggesting the formation of large aggregates of IR-820 and the stability of the PMID nanogel under biomimetic conditions; Figure S9: Evaluation of M2 macrophage polarization induced by IL-4 treatment. Macrophages were cultured in medium containing 200 ng/mL IL-4, and the expression of CD206, a surface marker for M2 macrophages, was evaluated by flow cytometry following immunostaining. IL-4 treatment upregulated the cell-surface expression of CD206, confirming successful polarization into the M2 phenotype.

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